

Infrastructure and Documentation



Chapter 2

Department of Computer Science & Informatics

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through quality, impact, and care.*

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UFS
NATURAL AND
AGRICULTURAL SCIENCES

Module Objectives

By the end of this module, you should be able to:

1. Describe the roles of various network and cabling equipment in commercial buildings and work areas
2. Maintain network documentation
3. Manage changes made to a network

Components of Structured Cabling

- ANSI/TIA-568 Commercial Building Wiring Standard
 - Also known as structured cabling
- The wiring standard describes the best way to install networking media to maximize performance and minimize upkeep
 - The principles apply no matter what type of media, transmission technology, or networking speeds are involved
- Structured cabling is based on a hierarchical design and assumes a network is based on the star topology

From the Demarc to a Workstation (1 of 11)

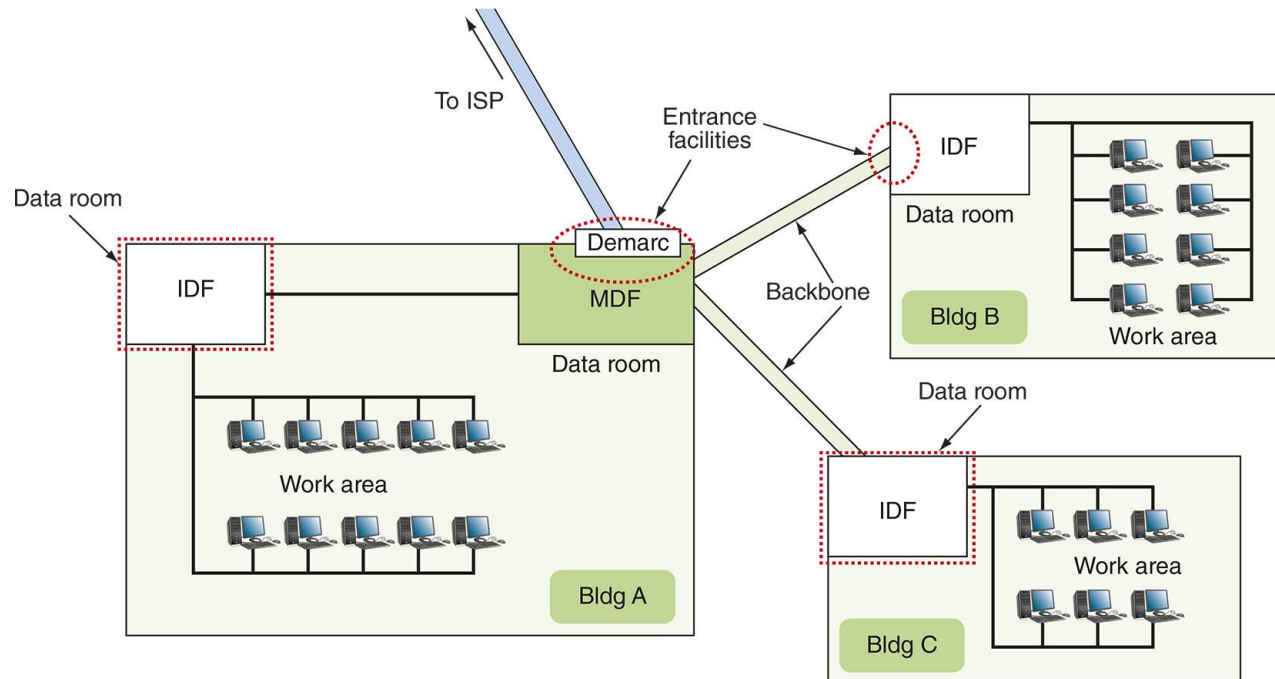


Figure 2-1 ANSI/TIA structured cabling in a campus network with three buildings

Figure 2-1 ANSI/TIA structured cabling in a campus network with three buildings

From the Demarc to a Workstation (2 of 11)

- Entrance Facility in Building A:
 - **EF (entrance facility)** – location where the incoming network (such as Internet) connects with the school or corporate network
 - **Demarc (demarcation point)** – the device that marks where a telecommunications service provider's network ends and the organization's network begins
 - **MDF (main distribution frame)** – the centralized point of interconnection for an organization's LAN or WAN (also called MC or main cross connect)
 - **Data room** – an enclosed space that hold network equipment (also called data closet, data center, equipment room, or telecommunications room)
 - **Rack** – holds various network equipment
 - **Patch panel** – a panel of data receptors which can be mounted to a wall or a rack
 - A patch panel provides a central termination point when many patch cables converge in a single location

From the Demarc to a Workstation (3 of 11)

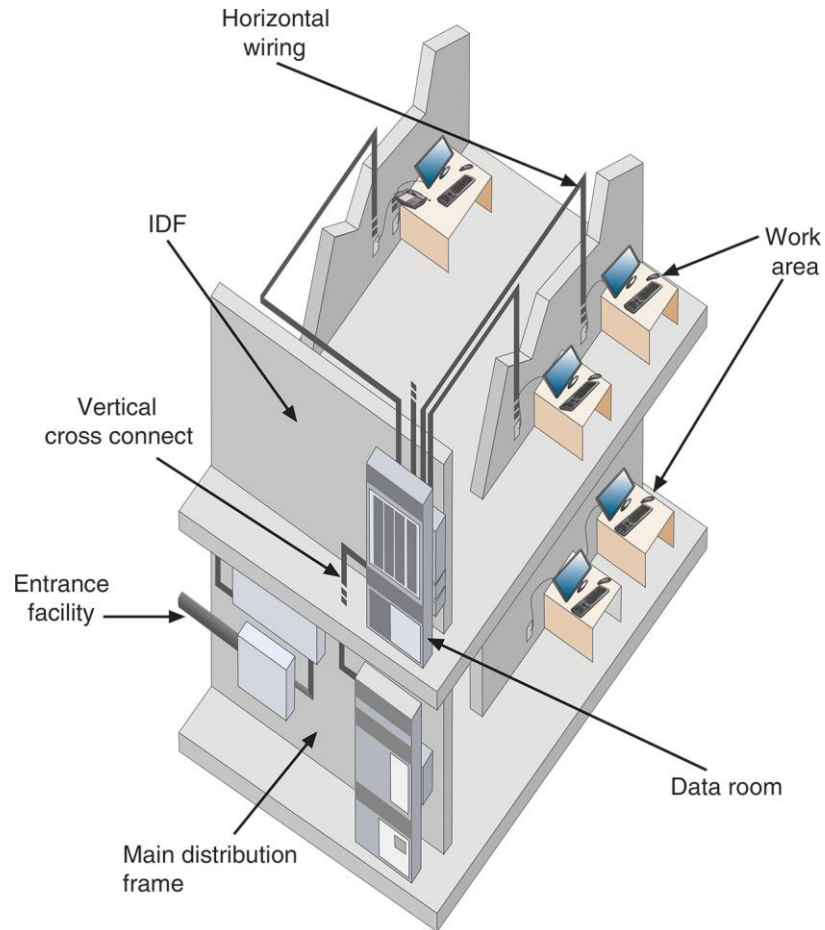


Figure 2-2 ANSI/TIA structured cabling inside a building

Figure 2-2 ANSI/TIA structured cabling inside a building

From the Demarc to a Workstation (4 of 11)



Figure 2-3 Demarc for Internet service to a campus network; this demarc is located inside a small data room and connects the incoming fiber signal from the ISP with the campus's Ethernet network

Figure 2-3 Demarc for Internet service to a campus network; this demarc is located inside a small data room and connects the incoming fiber signal from the ISP with the campus's Ethernet network

From the Demarc to a Workstation (5 of 11)



istock.com/thexfilephoto

Figure 2-5 Patch panel on rack

Figure 2-5 Patch panel on rack

From the Demarc to a Workstation (6 of 11)

- Entrance Facility in Building A (continued):
 - VoIP telephone equipment – VoIP (Voice over IP) is the use of any network to carry voice signals using TCP/IP protocols
 - In one or more data rooms you might find the following:
 - VoIP gateway
 - VoIP PBX
 - VoIP endpoints
- Data Room in Building B:
 - IDF (intermediate distribution frame) – provides an intermediate connection between the MDF and end-user equipment on each floor and in each building
- Work Areas in All Three Buildings:
 - Work area – encompasses workstations, printers, and other network devices
 - Wall jacks – the ANSI/TIA standard calls for each wall jack to contain at least one voice and one data outlet

From the Demarc to a Workstation (7 of 11)

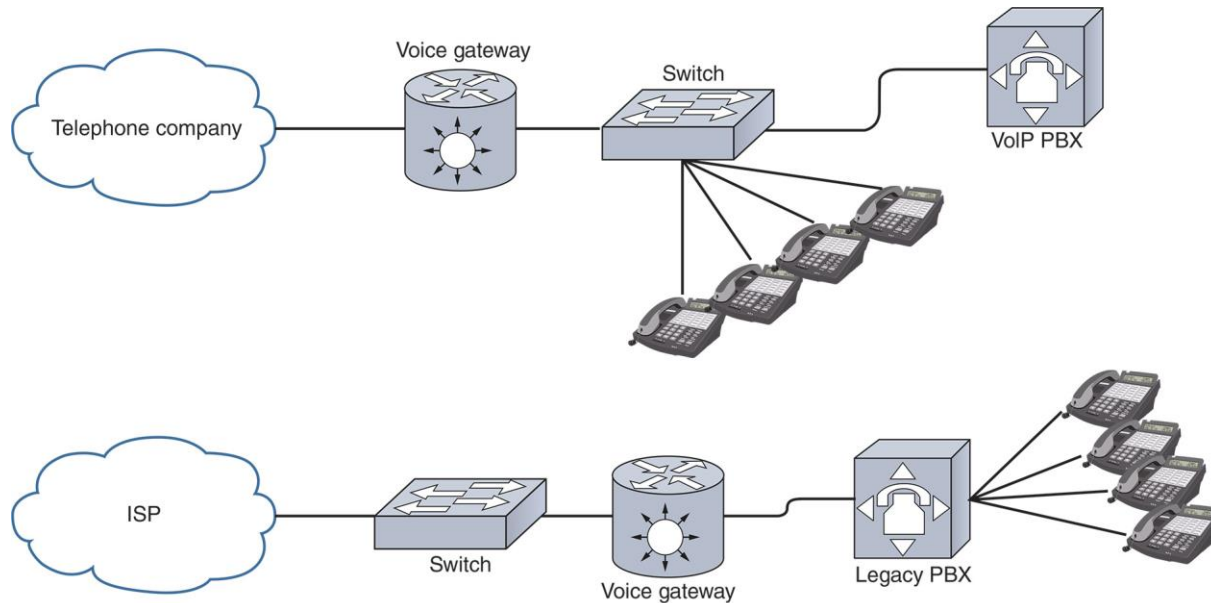


Figure 2-7 VoIP equipment can connect VoIP phones to an analog telephone line or an analog phone system to the Internet; there are pros and cons to each approach

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From the Demarc to a Workstation (8 of 11)

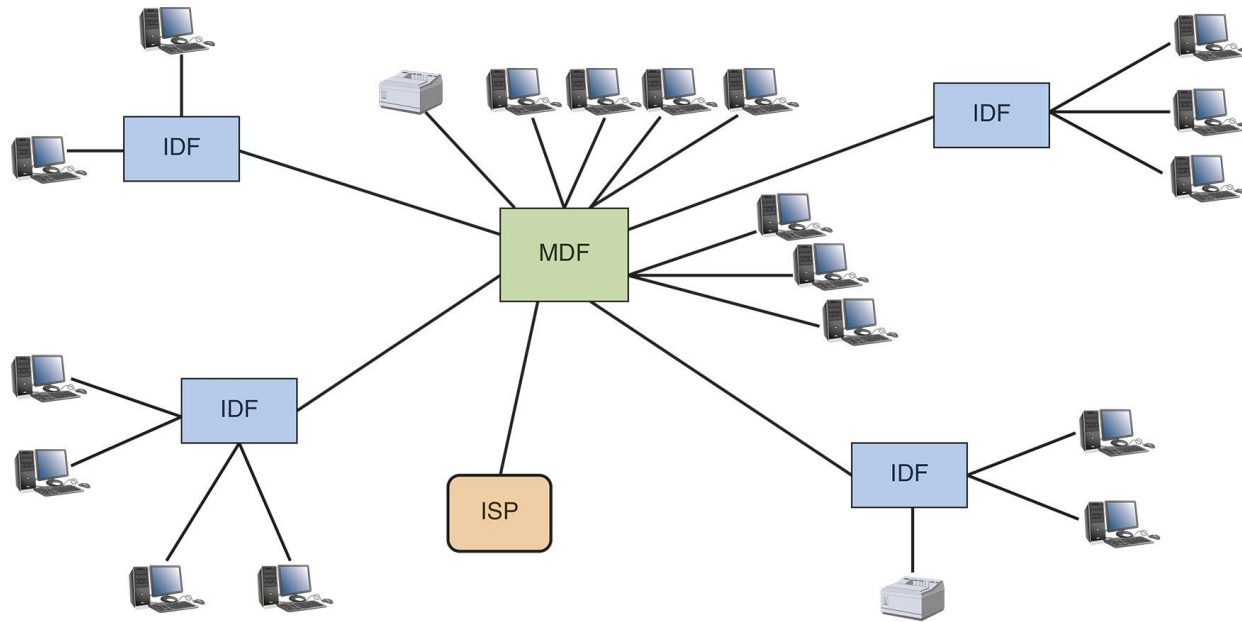


Figure 2-9 Workstations branching off IDFs that branch off an MDF create an extended star topology

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From the Demarc to a Workstation (9 of 11)

- Rack Systems
 - Racks come in two-post and four-post varieties (though six-post racks are also available)
 - Racks may be wall- or ceiling-mounted, freestanding on the floor, or bolted to the floor
 - Consider the following when purchasing racks:
 - Height – rack height is measured in rack units (RU or U) with the industry standard being 42U tall
 - Width – equipment racks come in a standard 19-inch frame (19 inches wide)
 - Depth – rack depths vary between manufacturers
 - In data centers containing multiple rows of racks, a hot aisle/cold aisle layout pulls cool air from vents in the floor or from low-lying wall vents into rows of racks (see Figure 2-14)

From the Demarc to a Workstation (10 of 11)



Figure 2-11 Open two-post racks and enclosed four-post racks

Figure 2-11 Open two-post racks and enclosed four-post racks

From the Demarc to a Workstation (11 of 11)

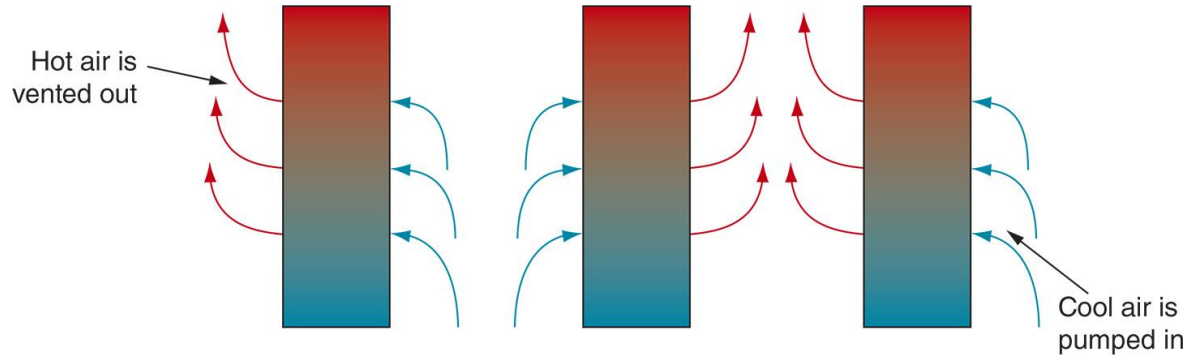


Figure 2-14 Hot aisle/cold aisle rack layout

Figure 2-14 Hot aisle/cold aisle rack layout

Cabling (1 of 5)

- Types of Cables
 - **Patch cable** – a relatively short length of cabling with connectors at both ends
 - **Horizontal cabling** – connects workstations to the closest data room and to switches housed in the room
 - **Backbone cabling** – consists of cables or wireless links that provide interconnection between the entrance facility and MDF and between MDF and IDFs
 - Many network problems are the result of poor cable installations
 - Pay close attention to the quality of cable connections and cable management

Cabling (2 of 5)

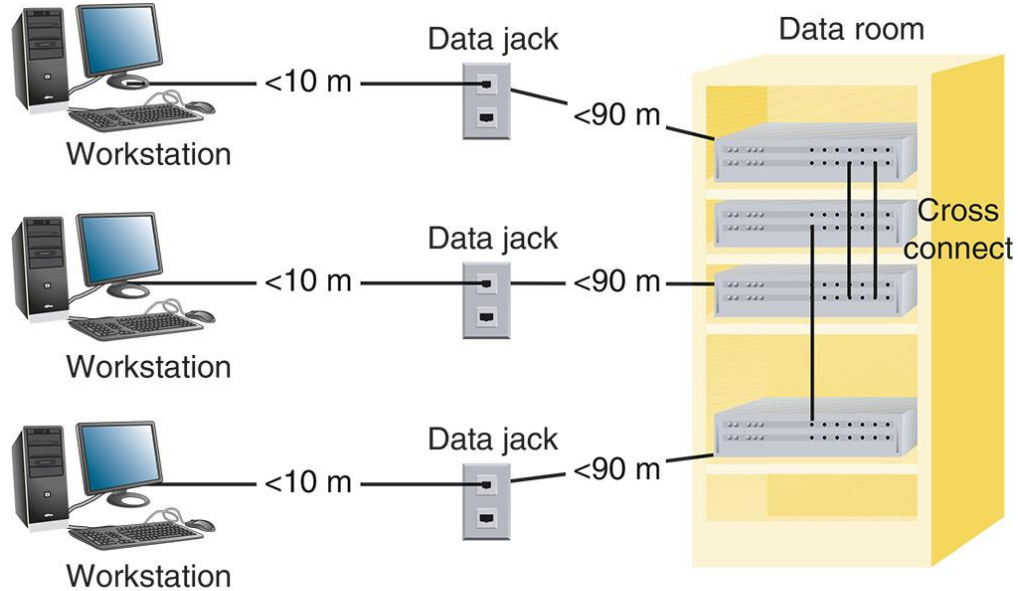


Figure 2-15 Horizontal cabling from a switch in a data room to workstations

Figure 2-15 Horizontal cabling from a switch in a data room to workstations

Cabling (3 of 5)

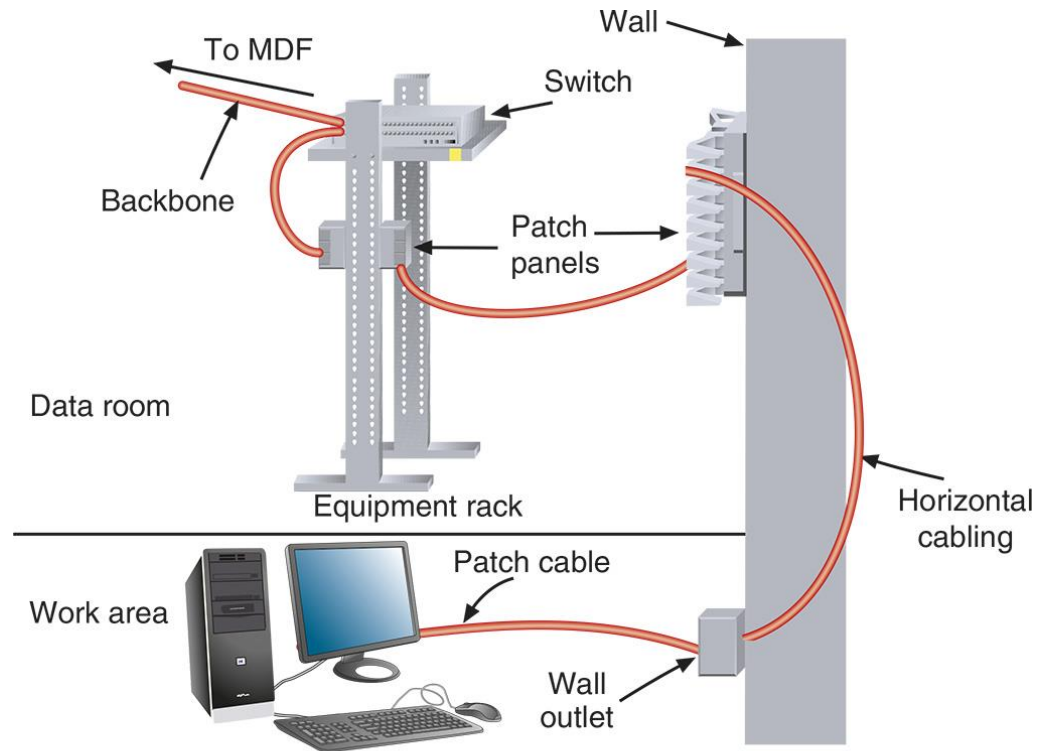


Figure 2-16 A typical UTP cabling installation

Figure 2-16 A typical UTP cabling installation

Cabling (4 of 5)

- Cable Management
 - **Termination** – when terminating twisted-pair cabling, don't leave more than 1 inch of exposed cable before a termination
 - **Bend radius** – do not exceed the cable's prescribed bend radius, which is the radius of the maximum arc into which you can loop a cable without impairing data transmission. The bend should be 4 times diameter of the cable.
 - **Continuity** – use a cable tester to verify that each cable segment transmits data reliably
 - **Loosely cinch cables**
 - **Cable coverings and conduits** – avoid laying cables across a floor and use cord covers if they must be exposed
 - **EMI sources** – install cable at least 3 feet away from fluorescent lights or other sources of EMI (electromagnetic interference)
 - **Plenum cabling** – if running cable in the plenum (the area above the ceiling tile or below subflooring), make sure the cable sheath is plenum-rated. A plenum-rated cable is coated with flame resistant jacket.

Cabling (5 of 5)

- Cable Management (continued)
 - **Grounding** - pay attention to *grounding* requirements
 - **Slack in cable runs**
 - **Cable trays** - use cable management devices such as *cable trays*, braided sleeving, and furniture grommets
 - **Patch panels** - use *patch panels* to organize and connect lines
 - **Company standards and inventory**
 - **Documentation**
 - Keep your cable plant documentation accessible
 - Label every data jack or port, patch panel and connector
 - Use color-coded cables for different purposes
 - Update your documentation as you make changes to the network

Monitoring the Environment and Security

- Data rooms are often serviced by HVAC systems separate from the rest of the building
- Specialized products can monitor the critical factors of a data closet's environment:
 - Unacceptable temperature
 - Humidity
 - Airflow conditions
- Industrial Control System (ICS) acquires real-time data from a physical system and manages the system or presents the data to humans to monitor and manage the system.
 - An example is a SCADA (supervisory control and data acquisition), an interface for widespread control that is used in different complex industrial appliances such power transmission, weather monitoring, etc.
- Every data room should be locked with only limited IT personnel having keys
 - Security cameras are suggested

Network Documentation

- Having up-to-date and detailed documentation of your network is essential for the following reasons:
 - Makes communication with coworkers more efficient
 - Speeds up troubleshooting efforts
 - Puts information at your fingertips when facing similar problems in the future

Network Diagrams (1 of 6)

- Network diagrams are graphical representations of a network's devices and connections
 - They may show physical layout, logical topology, IP address reserves, names of major network devices, and types of transmission media
- Network mapping - the process of discovering and identifying the devices on a network
 - Nmap – one of the most popular tools used for network mapping
 - Zenmap – Nmap's GUI option
- Cisco Systems set the standard for diagram symbols used to represent network devices

Network Diagrams (2 of 6)

```
PS C:\> nmap -T4 -F 192.168.2.1-254
Nmap scan engine for nmap.org ) at 2020-11-07 19:59 Eastern Standard Time
192.168.2.1
(ncy).
ports, 48 closed ports

192.168.2.120 (Tp-link Technologies)
(ncy).
ports

sn
ds
```

output in PowerShell using the `nmap` command

Figure 2-23 Nmap output in PowerShell using the `nmap` command

Network Diagrams (3 of 6)

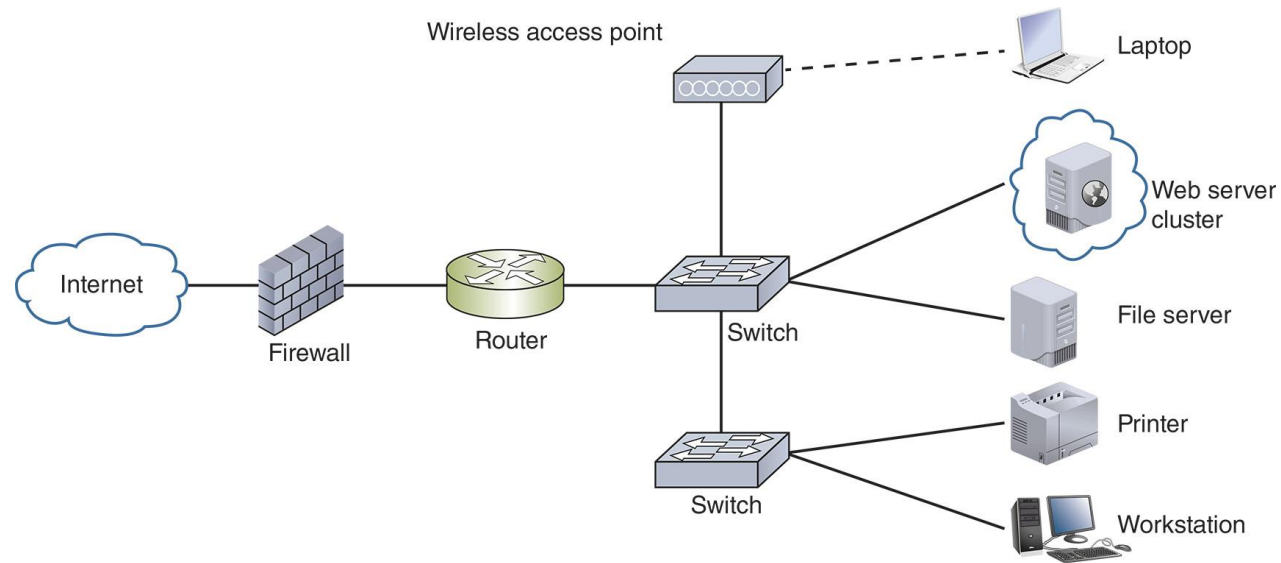


Figure 2-25 Network diagram using Cisco symbols

Figure 2-25 Network diagram using Cisco symbols

Network Diagrams (4 of 6)

- Network diagrams provide broad snapshots of a network's physical or logical topology
 - Useful for planning where to insert a new switch or determining how a particular router, gateway, or firewall interact
- Wiring schematic is a graphical representation of a network's wired infrastructure
 - In detailed form, it shows every wire necessary to interconnect network devices
- Rack diagram is a drawing that show devices stacked in a rack system

Network Diagrams (5 of 6)

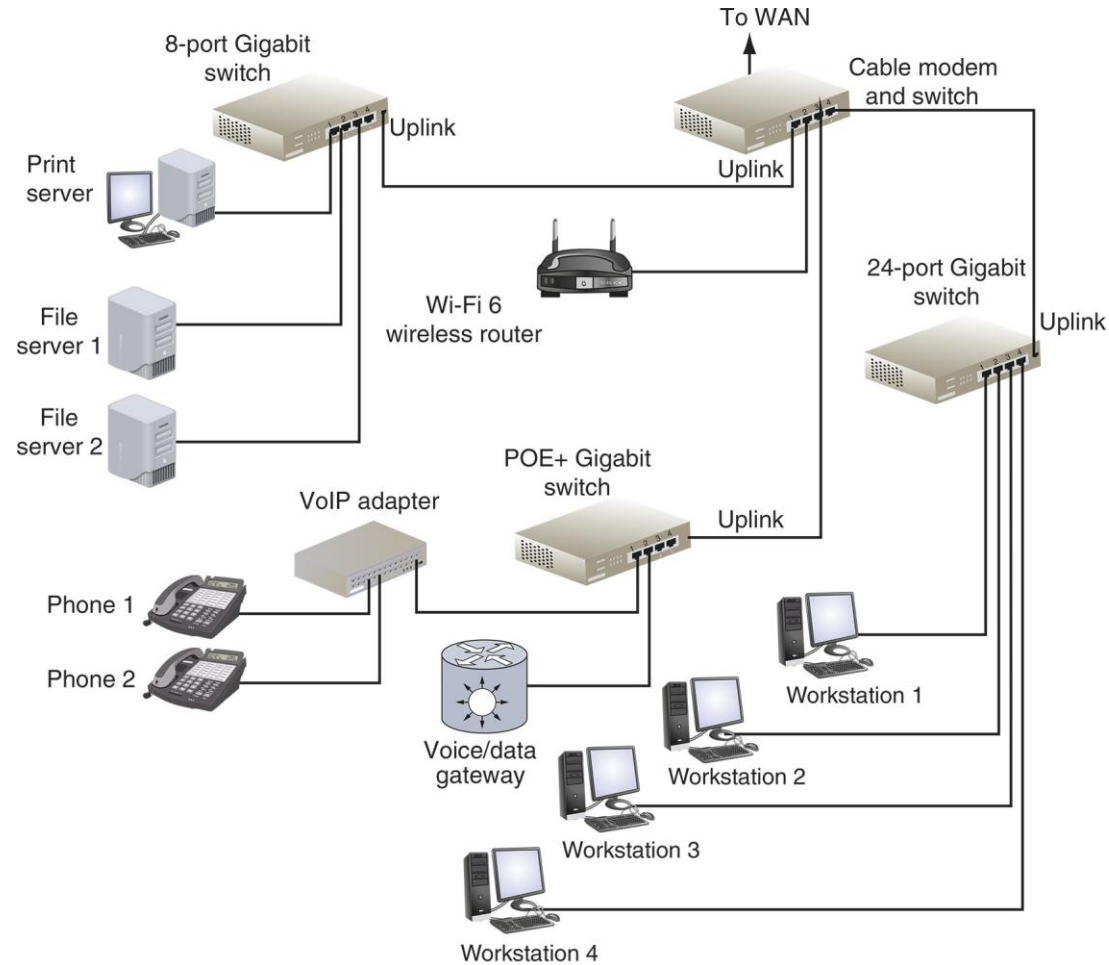


Figure 2-29 Wiring diagram

Figure 2-29 Wiring diagram

Network Diagrams (6 of 6)

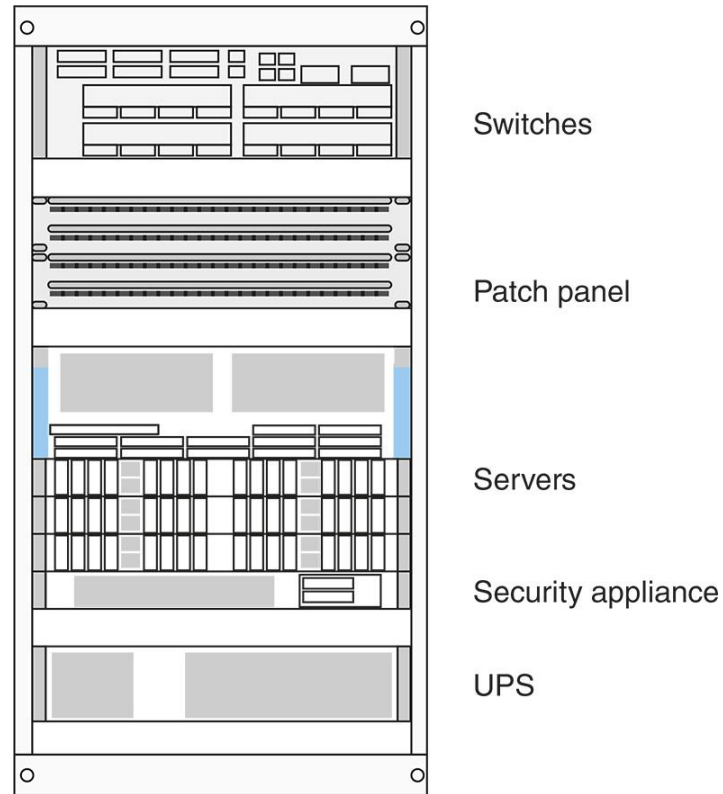


Figure 2-30 Typical devices installed on a rack

Figure 2-30 Typical devices installed on a rack

Operating Procedures

- Essential documentation:
 - Logical and physical connections on a network
 - Inventory management
 - IP address utilization
 - Vendors
 - Internal operating procedures
 - Policies
 - Standards
- Many corporations establish **SOPs (standard operating procedures)** to ensure consistency
- An organization could provide an internal website or database to hold this information

Inventory Management

- **System life cycle (SLC)** is the process of designing, implementing, and maintaining an entire network
- SLC includes:
 - Removal and disposal of outdated assets
 - Addition of compatible, updated devices
- **Inventory management** is the monitoring and maintaining of all assets that make up a network
 - Simplifies maintaining and upgrading a network
 - Provides network administrators with information about the costs and benefits of certain types of hardware or software

Labeling and Naming Conventions (1 of 3)

- Tips for labeling and naming conventions:
 - Use names that are as descriptive as possible
 - Only include fields that are essential in identifying the device
 - Don't overcomplicate the name with useless or redundant information
 - Use established naming conventions
 - Think big-picture-down-to-details
 - Consider any security risks from details included in your naming convention
 - Use color-coded cables and use cable tags to identify each cable's purpose
 - Label the ports and jacks that cables connect to
 - Where labels won't fit on the device, draw a simple diagram of each device that indicates how each port is used
 - Use labels that are durable and are designed to stick to plastic and metal

Labeling and Naming Conventions (2 of 3)

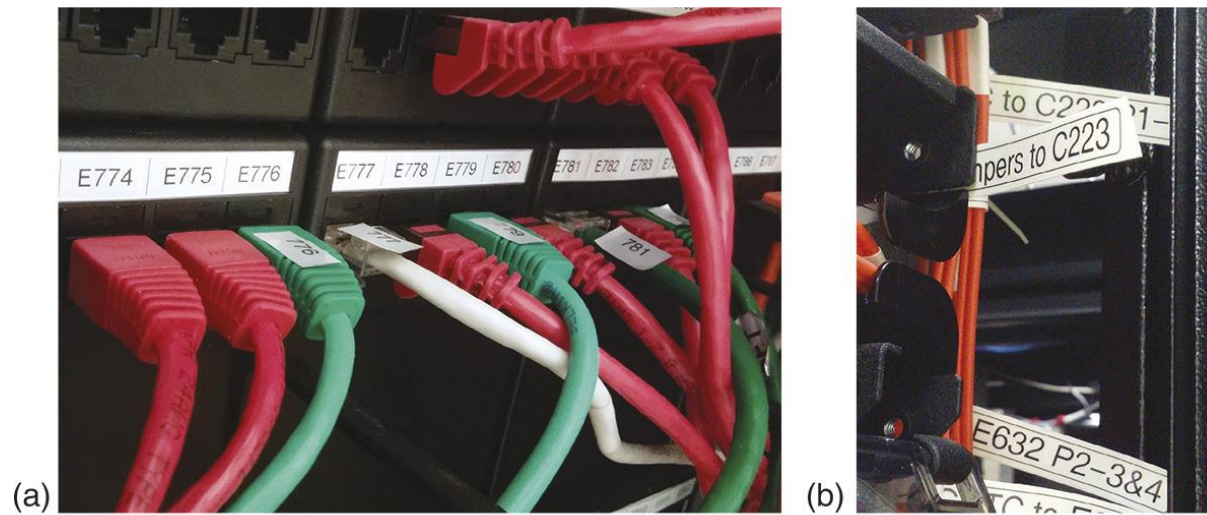


Figure 2-32 Labels on ports and tags on cables

Figure 2-32 Labels on ports and tags on cables

Labeling and Naming Conventions (3 of 3)

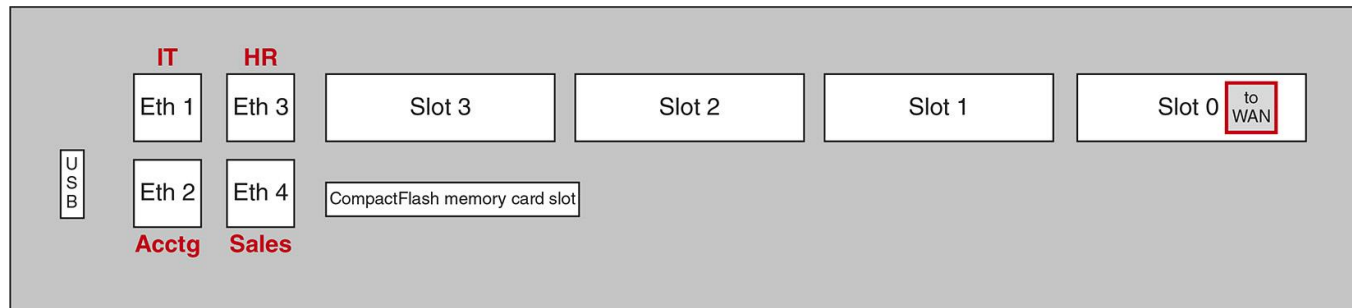


Figure 2-33 Simple diagram of a Cisco router with red labels identifying how five ports are used

Figure 2-33 Simple diagram of a Cisco router with red labels identifying how five ports are used

Business Documents (1 of 2)

- Standard business documents you may encounter:
 - RFP (request for proposal) is a request to vendors to submit a proposal for a product or service your company wants to purchase
 - MOU (memorandum of understanding) documents the intentions of two or more parties to enter into a binding agreement, or contract
 - SOW (statement of work) documents in detail the work that must be completed for a particular project
 - SLA (service-level agreement) is a legally binding contract or part of a contract that defines the aspects of a service provided to a customer
 - Example: the service provided by an ISP

Business Documents (2 of 2)

- Standard business documents you may encounter (continued):
 - MSA (master service agreement) is a contract that defines the terms of future contracts between parties
 - Examples include payment terms or arbitration arrangements
 - MLA (master license agreement) grants a license from a creator, developer, or producer to a third party for the purposes of marketing or sublicensing, or distributing the product to consumers

Change Management

- Managing change while maintaining network's efficiency and availability requires good planning
 - Network users need to know when to expect certain network resources to be unavailable
- Common software and hardware changes range from installing patches to replacing network backbone
- You may also need to know the change management documentation that might be required for an enterprise-scale network

Software and Hardware Changes (1 of 2)

- Four types of software changes:
 - Installation – new software to be installed on relevant devices and incorporated with network resources
 - Patch – a correction, improvement, or enhancement
 - Upgrade – a major change to a software package that enhances the functionality and features of the software
 - Rollback – also called backleveling or downgrading
 - Process of reverting to a previous version after attempting to patch or upgrade it
- Hardware changes could include adding new security cameras, replacing workstations, installing new printers, and upgrading hardware

Software and Hardware Changes (2 of 2)

- The same change management principles apply to any type of change:
 - Process all changes through the proper channels
 - Minimize negative impacts on business procedures
 - Plan thoroughly to maximize the chances of a successful change on the first attempt
 - Document each change throughout the process

Change Management Documentation

- Generally, the larger an organization, the more documentation required when making changes
- You may need to do the following:
 - Submit a change request document
 - Understand and follow the approval process
 - Follow project management procedures
 - Provide additional documentation that could include:
 - Network configuration
 - IP address utilization
 - Additions to the network
 - Physical location changes
 - Close the change

Summary

Now that the lesson has ended, you should be able to:

- Describe the roles of various network and cabling equipment in commercial buildings and work areas
- Maintain network documentation
- Manage changes made to a network